

Monte Carlo Radiative Transfer Beaming Functions for NICER Pulse-Profile Modeling

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ABSTRACT

NICER pulse-profile modeling relies on assumptions regarding the angular beaming of radiation emitted from neutron star surfaces. We present a Monte Carlo radiative transfer calculation of scattering-dominated neutron star atmospheres and extract angular beaming functions suitable for direct use in NICER pulse-profile modeling. Even modest deviations from isotropic emission are found to produce measurable changes in synthetic pulse profiles and pulsed fractions, highlighting the importance of realistic atmosphere models in precision neutron star studies.

1. INTRODUCTION

Pulse-profile modeling with NICER has enabled precise constraints on neutron star masses and radii (Bogdanov et al. 2019; Riley et al. 2019; Miller et al. 2019). A key systematic uncertainty in these analyses is the assumed angular distribution of emergent radiation from surface hot spots. Most current models adopt isotropic or parameterized limb-darkening laws. In this work, we explore angular beaming functions derived from first-principles Monte Carlo radiative transfer calculations.

2. MONTE CARLO RADIATIVE TRANSFER MODEL

We model a plane-parallel neutron star atmosphere using a photon-packet Monte Carlo approach (Lucy 1999). Photon packets are injected at the base of the atmosphere and propagate through a scattering-dominated medium. Optical depths are sampled using

$$\Delta\tau = -\ln(\xi),$$

where ξ is a uniform random deviate. Scattering directions are drawn from either isotropic or magnetically modified angular distributions.

3. ANGULAR BEAMING FUNCTIONS

Escaping photons are binned by $\mu = \cos\theta$ to construct normalized intensity distributions $I(\mu)$. For isotropic scattering, the emergent intensity approaches weak limb-darkening. Introducing magnetic anisotropy following Ho & Lai (2001) produces significant deviations from isotropy.

4. SYNTHETIC NICER PULSE PROFILES

We incorporate the Monte Carlo-derived beaming functions into a rotating hot-spot model with Schwarzschild light bending. Synthetic pulse profiles are generated for a range of viewing geometries. Compared to isotropic emission, anisotropic beaming modifies pulsed fractions by up to $\sim 15\%$, comparable to current NICER statistical uncertainties.

5. DISCUSSION AND CONCLUSIONS

Our results demonstrate that realistic radiative transfer effects can produce measurable changes in NICER pulse profiles. Even simplified Monte Carlo atmosphere models provide physically motivated beaming functions that improve upon ad hoc assumptions. These results highlight the importance of incorporating radiative transfer systematics in future NICER mass-radius analyses.

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